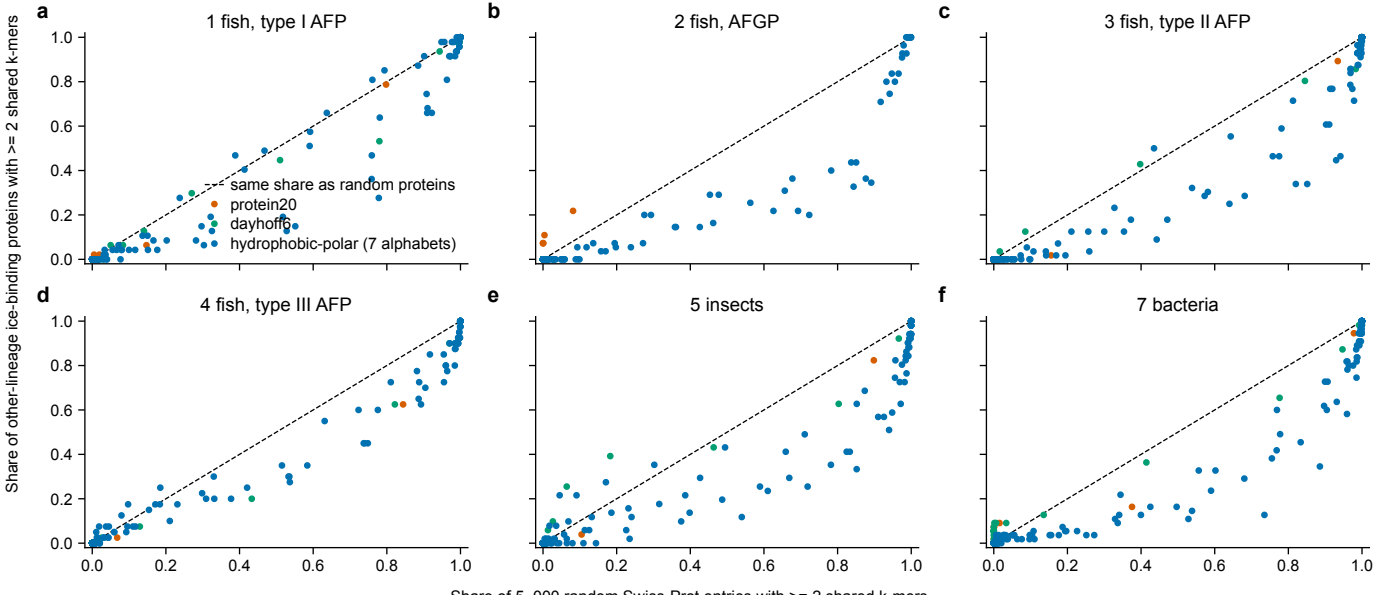TOOLS: shared distinct k-mers counted directly, k = 3 to 40, in 8 kmerseek alphabets; targets: reviewed Swiss-Prot 2026_03
HYPOTHESIS: some alphabet and k lets a query share k-mers with ice-binding proteins of other lineages far more often than with random proteins (points well above the diagonal)
CONCLUSION: with at most 1% of random proteins reachable, the most any query reaches is 6 proteins of another lineage; the hydrophobic-polar points sit on or below the diagonal